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THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

MEASUREMENTS AND INSTRUMENTATION FOR AEROSPACE

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **5 X 2=10**

			Blooms Level	CO	Marks
1.	**a.**	Differentiate between active and passive sensors with an example	L2	CO1	**2**
	b.	Draw the circuit diagram of load cell and explain its functionality	L2	CO2	**2**
	c.	List out all air and gyroscopic data instruments used in aircraft	L2	CO3	**2**
	d.	Differentiate between Surface contact type and immersion type thermocouples	L2	CO4	**2**
	e.	Explain how RPM is sensed in aircraft using a Hall Effect Sensor.	L2	CO4	**2**
		PART-B		**6 X 5=30**	
2.	**a.**	Classify various errors observed in measurements and differentiate them on their basis of comparison.	L2	CO2	**5**
		OR			
	b.	Discuss in detail the static and dynamic characteristic of instrumentation	L2	CO1	**5**
3.	**a.**	List the fundamental principles of the mechanical gyroscope with the help of neat sketch. Also explain the mechanism of powering gyro.	L2	CO3	**5**
		OR			
	b.	Describe the working principle of fiber optic gyro with a neat diagram.	L2	CO3	**5**
4.	**a.**	Discuss the characteristics required for the cockpit LCD panels in detail with neat sketches.	L2	CO3	**5**

		OR			
	b.	What is static position error and how do you determine it and discuss strategy adopted to reduce the static position error with neat sketch.	L2	CO2	**5**
5.	**a.**	Illustrate general LVDT assembly and explain its working principle with the circuit diagram.	L2	CO3	**5**
		OR			
	b.	Discuss the ISO 2631-1 standard specifications to measure and evaluate whole-body vibration.	L2	CO4	**5**
6.	**a.**	With the help of neat sketch, explain the components of a turbine engine used in aircraft and label each of the components. Alongside discuss on the various temperature sensors used in engine temperature monitoring.	L2	CO4	**5**
		OR			
	b.	**i.** LVDT stands for _____ **ii.** Mach number _____ **iii.** MFD stands for _____ **iv.** RTD sensor is used to measure _____ **v.** MEMS stands for _____	L1	CO1-CO5	**5**
7.	**a.**	A pitot static tube is mounted on an aircraft travelling at a speed 300 kmph against a wind velocity of 20 kmph. If the specific weight of air is 12 N/m3 determine the pressure difference the instrument will register.	L3	CO2	**5**
		OR			
	b.	Illustrate the difference between cambered and symmetric airfoil with neat sketch. Give the nomenclature of the airfoil and explain what characteristics will change in both of the airfoils.	L2	CO3	**5**
		PART-C		**6 X 10=60**	
8.	**a.**	A K thermocouple produces a voltage which is measured by the potentiometer as 25mV. Determine the temperature T when the Reference Junction isothermal block is indicated	L3	CO4	**10**

		by a thermistor as 0°C. Use the Seebeck coefficient for 20°C.			
		OR			
	b.	Distinguish between static and dynamic errors with examples to each of these types of errors.	L2	CO1	**10**
9.	**a.**	A gyroscope spins with its tip on the ground and is spinning with negligible frictional resistance. The disk of the gyroscope has mass 0.3 kg and is spinning at 20 rev/s. Its center of mass is 5.0 cm from the pivot and the radius of the disk is 5.0 cm. What is the precessional period of the gyroscope?	L3	CO3	**10**
		OR			
	b.	Discuss in brief the following **i.** Inductor Pressure Transmitter **ii.** Pressure switches **iii.** MEMS Pressure Sensors	L2	CO4	**10**
10.	**a.**	Detail about HPT-32 aircraft with neat sketch, also list the technical specifications, display devices in the cockpit and its instrumentation.	L2	CO2	**10**
		OR			
	b.	Construct block diagram of the Gyroscopic data instruments in aircraft with the help of gyroscope.	L2	CO3	**10**
11.	**a.**	Define the following with the help of neat sketches **i.** Angle of attack **ii.** V_{stall} **iii.** Side slip **iv.** Pressure, temperature and density at MSL **v.** Compressibility	L1	CO1-CO5	**10**
		OR			
	b.	State principle of pulse code modulation, explain the PCM process with the help of the block diagram	L2	CO5	**10**
12.	**a.**	A parallel plate capacitor has square plates of side 5 cm and	L3	CO4	**10**

		separated by a distance of 1 mm. (a) Calculate the capacitance of this capacitor. (b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates? (The value of ε_o = 8.85 x 10-12 Nm^2C^{-2})			
		OR			
	b.	1. Discuss about any of the sensor of your choice, with neat sketch, circuit diagram, working principal, and an application to the aircraft. 2. Calculate the gauge factor of a strain gauge, if the value of resistance is 152 ohms, which changes by 5 ohms for 500 micro strain.	L3	CO1	**10**
13.	**a.**	State the principle of operation of FDR and parameters recorded in FDR. Also list the FDR survivability standards.	L2	CO5	**10**
		OR			
	b.	A resistance temperature detector (RTD) arranged in a bridge with each arm 100 ohm and volt supply. If the temperature of the sensor changed such that the meter indicates 0.569 volt, the material has temperature coefficient of 0.0039/°C. The resistance at 0°C is 100 Ω. Find the sensor temperature.	L3	CO4	**10**